

Project Problem Statement: Optimizing Urban Mobility & Ride Fulfillment

Context

As one of India's leading urban mobility platforms, Rapido processes hundreds of thousands of ride requests daily across various vehicle segments, including Bikes, Autos, and Cabs, within highly congested metropolitan hubs like Bengaluru. Maintaining marketplace equilibrium—ensuring high ride completion rates, minimizing wait times (TAT), and maximizing revenue—requires continuous monitoring of transaction data.

The Challenge

Despite steady ride demand, the platform experiences friction points that degrade the user experience and leak revenue. Specifically:

- **Ride Cancellations & Incompletes:** A significant volume of bookings fail due to customer cancellations, driver rejections, or vehicle breakdowns, leading to sunk operational costs and frustrated users.
- **Fleet Utilization Imbalances:** Ride distances, driver ratings, and customer ratings vary heavily across vehicle categories, masking inefficiencies in how different tiers (e.g., Bike vs. Sedan) perform.
- **Revenue Volatility:** Fluctuations in booking values during high-demand windows (weekends and cricket match days) present an unoptimized surge and pricing strategy.

Objective

The objective of this project is to analyze a comprehensive transaction dataset of **100,000 bookings** over a one-month period in Bengaluru using **SQL** and **Power BI**. By auditing data integrity, tracking Key Performance Indicators (KPIs), and building an interactive analytical dashboard, this project aims to:

1. Identify the primary operational root causes behind customer and driver cancellations.
2. Segment vehicle and customer performance to uncover high-value cohorts and underperforming asset classes.
3. Provide actionable, data-driven recommendations to improve ride fulfillment rates, stabilize driver retention, and optimize revenue streams during peak demand cycles.